Diet, infection and wheezy illness: lessons from adults.

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An increase in asthma and atopic disease has been recorded in many countries where society has become more prosperous. We have investigated two possible explanations: a reduction in childhood infections and a change in diet. In a cohort of people followed up since 1964, originally selected as a random sample of primary school children, we have investigated the relevance of family size and the common childhood infectious diseases to development of eczema, hay fever and asthma. Although membership of a large family reduced risks of hay fever and eczema (but not asthma), this was not explained by the infections the child had suffered. Indeed, the more infections the child had had, the greater the likelihood of asthma, although measles gave a modest measure of protection. We have investigated dietary factors in two separate studies. In the first, we have shown the risks of bronchial hyper-reactivity are increased seven-fold among those with the lowest intake of vitamin C, while the lowest intake of saturated fats gave a 10-fold protection. In the second, we have shown that the risk of adult-onset wheezy illness is increased five-fold by the lowest intake of vitamin E and doubled by the lowest intake of vitamin C. These results were supported by direct measurements of the vitamins and triglycerides in plasma. We have proposed that changes in the diet of pregnant women may have reflected those observed in the population as a whole and that these may have resulted in the birth of cohorts of children predisposed to atopy and asthma. The direct test of this is to study the diet and nutritional status of a large cohort of pregnant women and to follow their offspring forward. This is our current research.